

Qatar-2b

R-2036

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_120511 · created 2026-08-08T01:46:41+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456058.83064 +/- 0.00043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.127 +/- 0.031
Transit depth (Rp/Rs)^2	1.61 +/- 0.78 [%]
Orbital Inclination (inc)	85.83 +/- 1.65
Airmass coefficient 1 (a1)	0.963 +/- 0.022
Airmass coefficient 2 (a2)	0.0172 +/- 0.0087
Scatter in the residuals of the lightcurve fit is	2.23 %
Best Comparison Star	#2 - [356, 328]
Optimal Aperture	3.17
Optimal Annulus	11.48
Transit Duration (day)	0.0669 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.127 $\pm$ 0.031 vs 0.16327 (deviation 1.17 σ)
Duration fit vs published	0.0669 d vs 0.0767 d (deviation 1.46 σ)
Mid-time O-C	0.9 min
Depth SNR	4.1
Residual scatter	2.23 %

Light curve

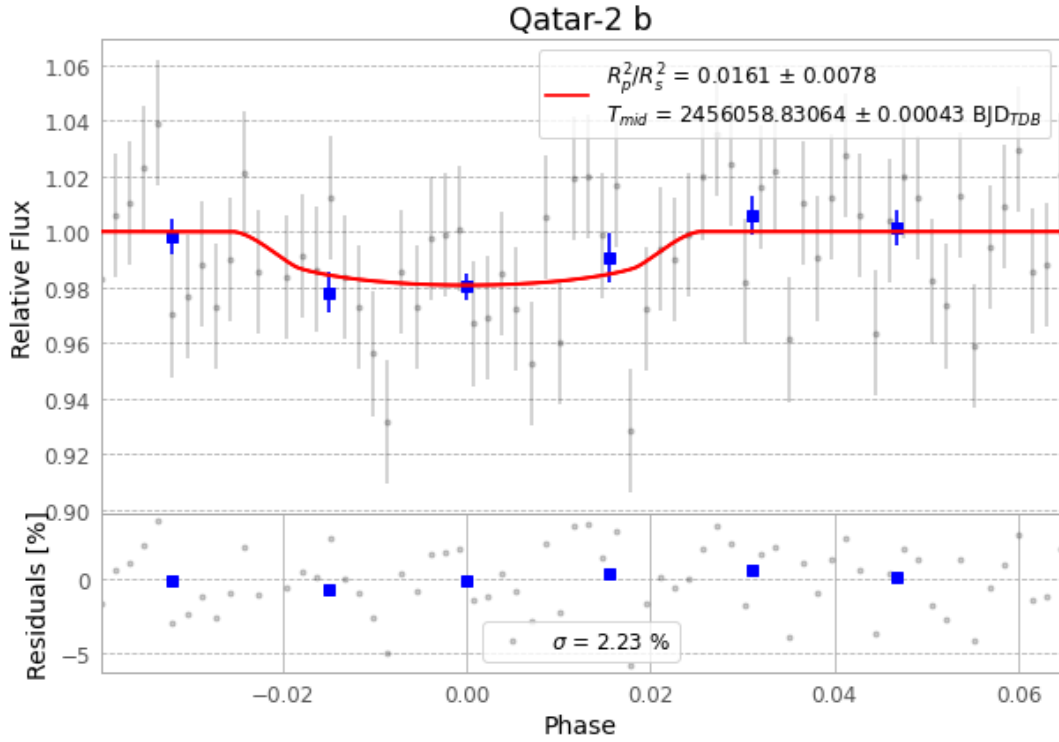


Figure 1 — FinalLightCurve_Qatar-2 b_2012-05-10.png (SHA-256 9794ad51a7c6157e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 309], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b798eb155f89f360...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

68 FITS frames (45 MB), Qatar-2120511063014.FITS → Qatar-2120511095112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-120511.log (306152f16ee2...), FinalLightCurve_Qatar-2 b_2012-05-10.png (9794ad51a7c6...), FinalLightCurve_Qatar-2 b_2012-05-10.csv (8cebed236c69...)

Compute resources

Wall time 102.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 01:46Z.